Lake Arden water-quality monitoring

Summer 2027 · interim report of the lake management group

Synthetic sample: the lake, the monitoring stations and all measurements in this report are invented.

1 Summary

Lake Arden has been sampled every week since 1 June 2027 at six stations, from the inflow of Mill Brook in the north to the outflow weir in the south. This interim report covers the samples taken up to the end of August. It describes the algae indicator chlorophyll-a, the station averages of temperature, water clarity and phosphorus, and the oxygen conditions near the surface at station B.

The summer was warmer than the long-term average at most stations. Algae concentrations increased through the season, most strongly near the brook inflow, where the nutrient load is highest. Water clarity was lowest at the shallow stations in the north-east bay.

Dissolved oxygen at station B stayed above the target minimum until early August, fell below it for three weeks and recovered after a storm on 20 August mixed the water. No fish deaths were reported.

The management group will decide in October whether the planned reduction of the phosphorus load from the brook catchment should be brought forward. The final report, with the autumn samples, follows in January 2028.

Monitoring stations

A outflow weir · B central basin, deepest point · C Mill Brook inflow · D western shore · E north-east bay · F boat harbour

Samples are taken between 09:00 and 11:00 from a boat at 1 m depth, except where the table gives a different depth. All analyses were made by the same accredited laboratory.

2 Chlorophyll-a

Chlorophyll-a is used as a measure of the amount of algae in the water. Figure 1 shows the monthly mean of the weekly surface samples at the four stations in the open lake. Stations E and F are left out because several samples could not be taken when the bay was closed for works on the harbour wall.

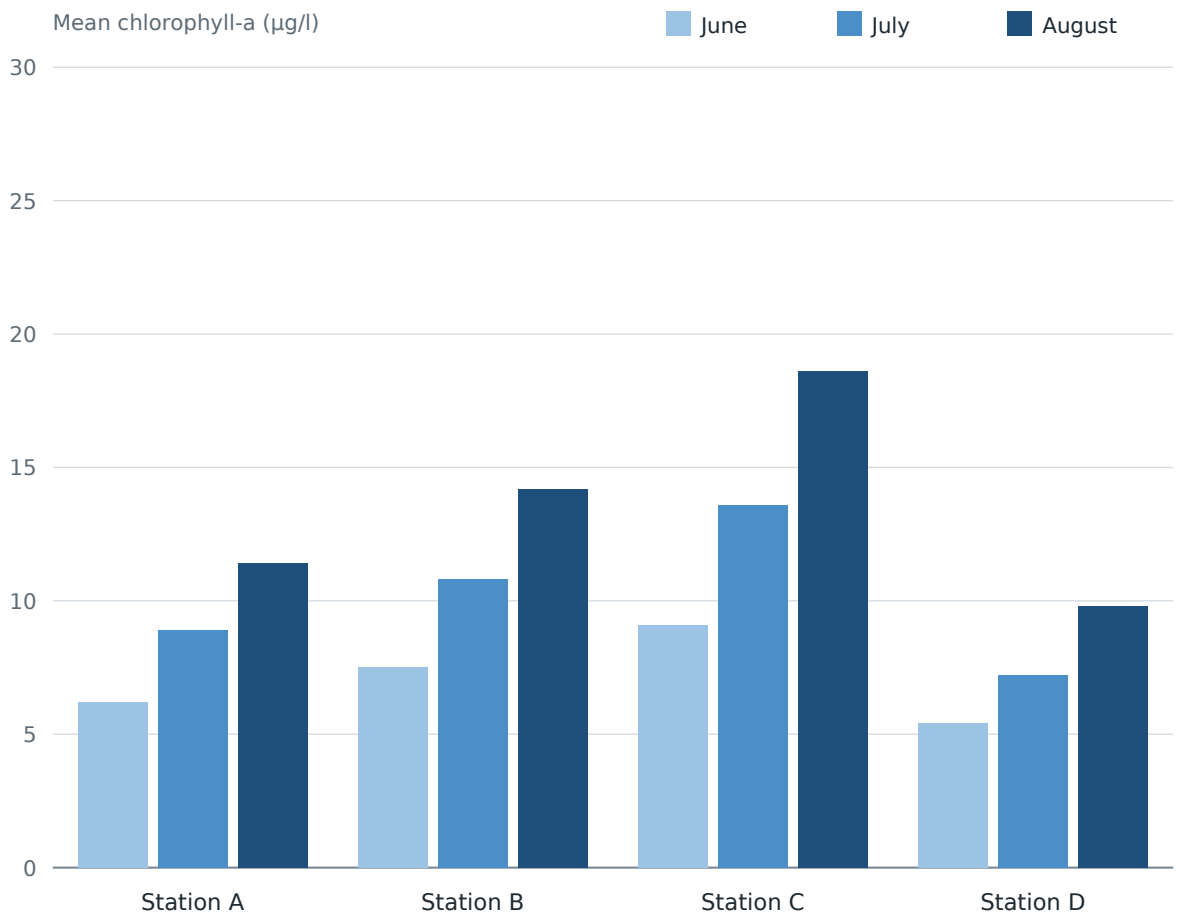


Figure 1. Monthly mean chlorophyll-a concentration at stations A–D, surface samples, June to August 2027.

Concentrations rose at all four stations through the summer. The increase was largest at station C near the inflow of Mill Brook, which carries most of the phosphorus that enters the lake. The central basin (station B) followed the same pattern at a lower level, and the western shore (station D) remained the clearest part of the open lake.

An algae bloom warning is issued when chlorophyll-a stays above the warning level for two consecutive weeks. This did not happen at any station this summer, but station C came close to it in the last week of August.

3 Station summary

Table 1 gives the averages over all samples from June to August. The temperature anomaly is the difference from the mean of the summers 2012–2026 at the same station; a negative value means the water was cooler than usual.

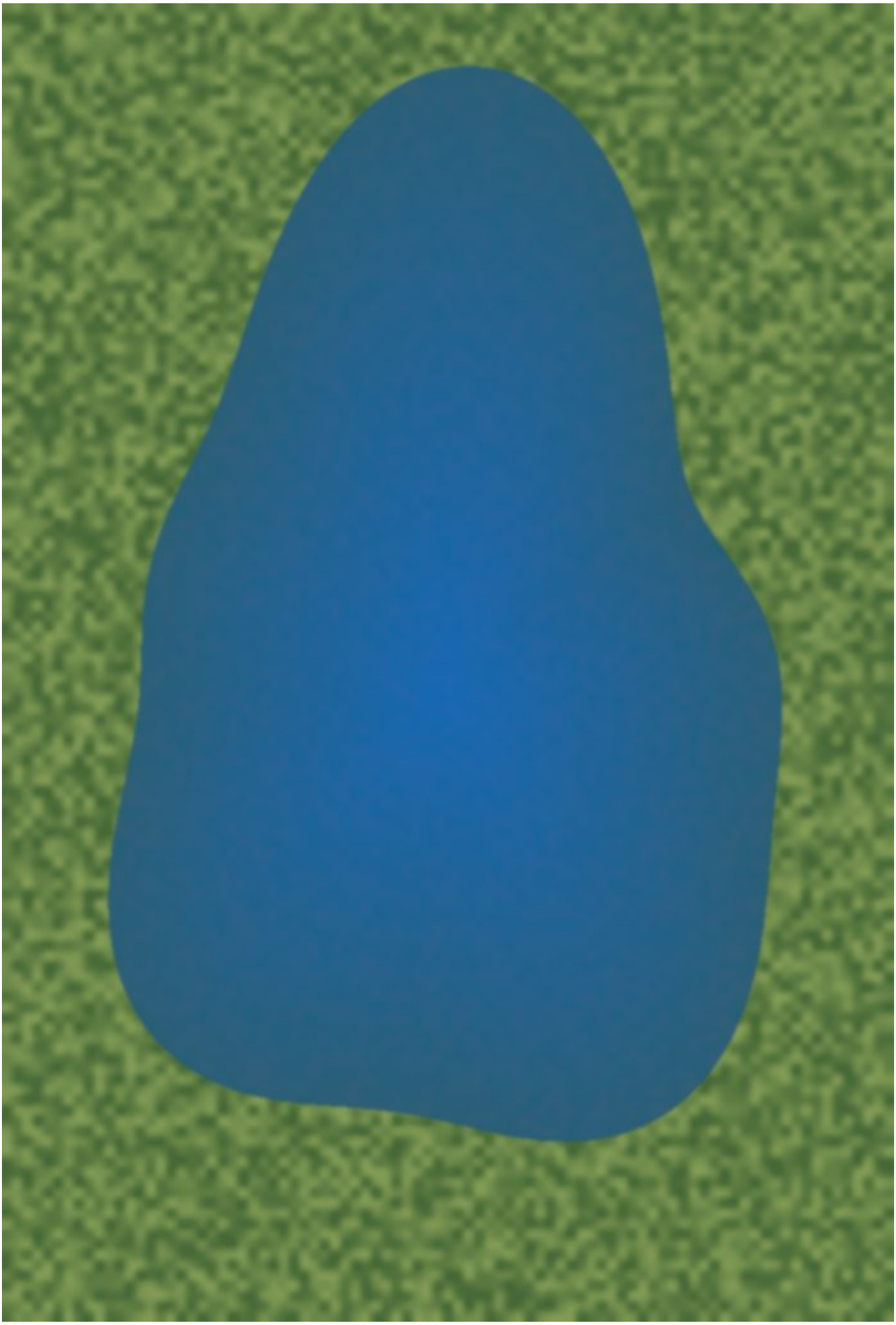
Station	Depth (m)	Mean temp. (°C)	Anomaly (°C)	Secchi depth (m)	Total P (µg/l)
A	12.5	19.8	+0.6	3.4	18
B	21.0	19.1	−0.2	3.9	15
C	6.5	20.7	+1.3	1.8	31
D	15.0	19.4	−0.4	3.1	22
E	4.0	21.2	+1.6	1.2	38
F	9.5	20.1	+0.9	2.6	27

Table 1. Station averages, June to August 2027. Shaded cells: total phosphorus above the target of 35 µg/l.

The shallow stations C and E warmed most and had the lowest water clarity. Both are close to the brook inflow and to the reed beds of the north-east bay, where fine sediment is stirred up by wind and boats.

Total phosphorus decreased from north to south. The outflow weir (station A) had the lowest value of the open-lake stations, which is consistent with phosphorus settling out with algae and sediment on the way through the lake.

The depth column gives the depth of the water at the sampling point, not the sampling depth. Secchi depth is the depth at which a white disc lowered from the boat can no longer be seen.



4 Dissolved oxygen

Oxygen is measured with a calibrated probe at 1 m depth in the central basin. Figure 2 shows the weekly values together with the target minimum of 6 mg/l that applies to lakes with a cold-water fish population.

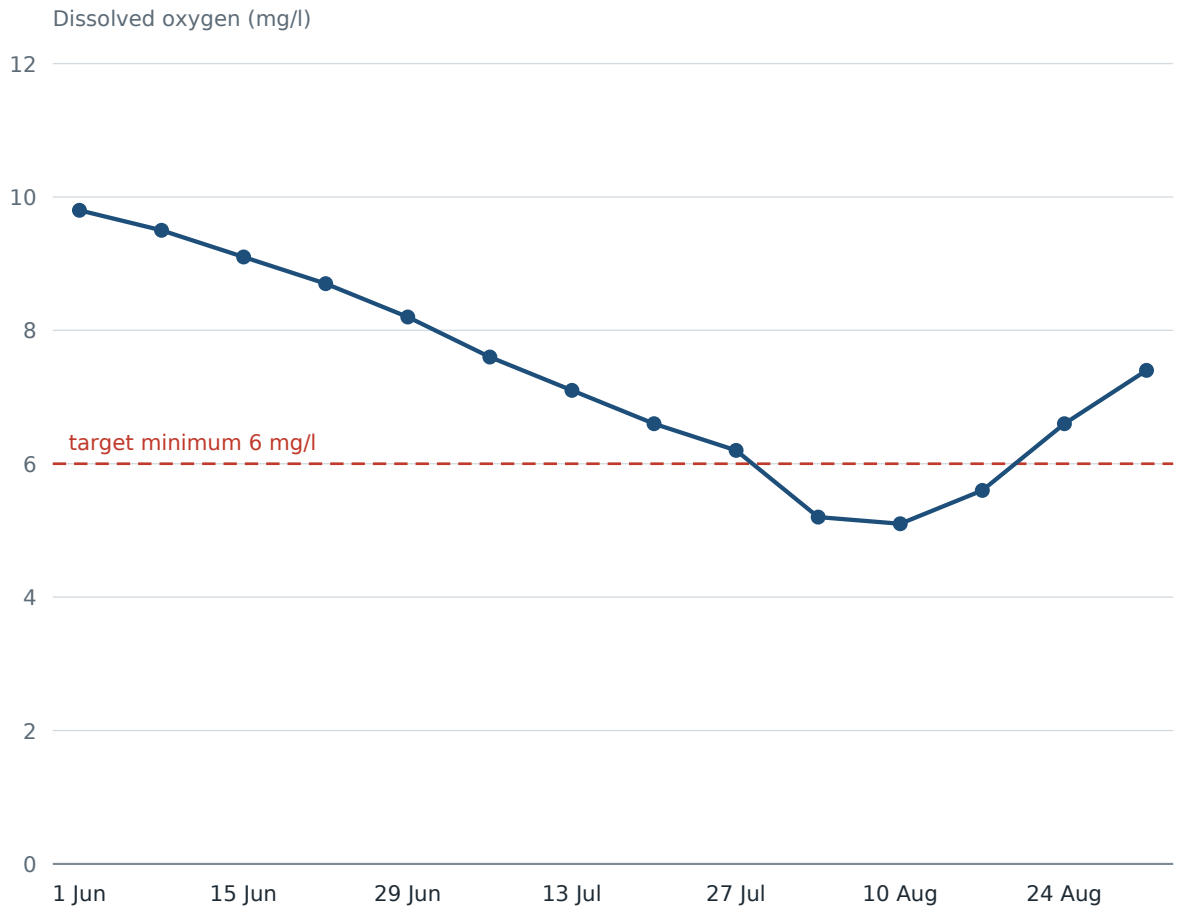


Figure 2. Dissolved oxygen at 1 m depth, station B, weekly samples from 1 June to 31 August 2027.

The values fell steadily through July as the water warmed and algae decomposed in the deeper layers. They were below the target minimum in the three weeks from 3 to 17 August and returned above it after the storm of 20 August mixed the water column.

The probe was checked against laboratory titration every fortnight. All checks were within the tolerance of the method, so no values had to be removed.

5 Conclusions and next steps

The summer results confirm the pattern of earlier years: the northern part of the lake, fed by Mill Brook, carries the highest nutrient and algae levels, and the central basin is at risk of low oxygen in warm, calm weeks.

The management group recommends three steps for the coming year:

• continue weekly sampling at all six stations until the end of October;

• add a second oxygen probe at 8 m depth in the central basin, so that the deeper water can be followed;

• ask the farms in the Mill Brook catchment to report fertiliser use for the 2028 season.

The cost of the extra probe and the additional laboratory work is covered by the existing monitoring budget. Comments on this interim report can be sent to the secretary of the lake management group until 30 November 2027.